

UNDERSTANDING HACKING

An overview of hacking concepts, types, and cybersecurity defenses

WHAT IS HACKING?

UNDERSTANDING HACKING

- Unauthorized access to computer systems or networks
- Exploits vulnerabilities in software, hardware, or human behavior
- Can be malicious, ethical, or exploratory in nature
- Modern hacking often targets data, finances, or infrastructure



Hacking is the act of compromising digital systems, with intent ranging from criminal to protective.

TYPES OF HACKERS

- Black hat: malicious actors seeking personal gain or damage
- White hat: ethical hackers authorized to test and improve security
- Grey hat: operate without permission but may report vulnerabilities
- Script kiddies: inexperienced users relying on pre-made tools



COMMON ATTACK VECTORS

- Phishing: deceptive emails or messages stealing credentials
- Malware: viruses, ransomware, spyware delivered via downloads
- SQL injection: manipulating database queries through input fields
- Brute force: automated password guessing attacks



Attackers exploit technical flaws and human trust through diverse, evolving methods.

ETHICAL HACKING & PENETRATION TESTING

UNDERSTANDING HACKING

- Authorized simulated attacks to identify weaknesses
- Follows frameworks like OWASP, NIST, or PTES
- Includes reconnaissance, scanning, exploitation, and reporting
- Certifications: CEH, OSCP, GPEN validate professional skills



Ethical hacking proactively finds vulnerabilities before malicious actors do.

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DEFENSIVE CYBERSECURITY MEASURES

- Multi-factor authentication prevents credential theft
- Regular patching closes known software vulnerabilities
- Network segmentation limits lateral movement after breach
- Security awareness training reduces phishing success rates



Layered defenses combining technology, process, and people reduce risk significantly.

KEY TAKEAWAYS

- Hacking spans a spectrum from criminal to protective activity
- Understanding attack methods enables better defense
- Ethical hacking is a legitimate, certified profession
- Security requires continuous adaptation, not one-time fixes



THANK YOU

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